

Product Name: 3.5 inch TFT LCD
Model No. :
Version : V0
Date : 2025-07-17

客户签回 (APPROVED BY CUSTOMER)	
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制表 (Prepared by)	检查 (Checked by)	审核 (Approved by)
农工	易工	易工

Revision Notice	Description	Page	Rev. Date
1.0	First version	--	2025/07/17

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1. GENERAL INFORMATION

Item	Contents	Unit
LCM outline dimension	76.84(W) * 63.84(H) * 3.27(T)	mm
Module size	3.5	inch
Resolution	640(RGB) *480	-
Active area	70.08 (W) *52.56 (H)	mm
Pixel pitch	-	um
LCD type	IPS	-
LCD part No.	-	-
Viewing direction	ALL	-
LCM driver IC	HX8394F	-
Touch panel	-	-
LCM interface	MIPI	-
Color depth	16.7M	-
Backlight type	LED*12	-
POL type	-	-

Model Number details:

KLD35176BVIC

KLD Company Name

35 Panel Size

176 LCM model code(Internal control)

B Panel maker

V Resolution

I Connector type

C Version

2. INTERFACE CONNECTION :

Pin No.	Symbol	Functional
1	GND	Connection Ground
2	MIPI_D0N	Differential data0 signals
3	MIPI_D0P	Differential data0 signals
4	GND	Connection Ground
5	MIPI_D1N	Differential data1 signals
6	MIPI_D1P	Differential data1 signals
7	GND	Connection Ground
8	MIPI_CLKN	Differential clock signals
9	MIPI_CLKP	Differential clock signals
10	GND	Connection Ground
11	NC	DUMMY
12	NC	DUMMY
13	GND	Connection Ground
14	NC	DUMMY
15	NC	DUMMY
16	GND	Connection Ground
17	REST	Reset pin. Setting either pin low initializes the LSI.
18	NC	DUMMY
19	NC	DUMMY
20	NC	DUMMY
21	NC	DUMMY
22	IOVCC	Power supply for interface pins. (1.65V-3.3V)
23	GND	Connection Ground
24	VSN	Negative power supply for source driver and step-up circuits. Connect a 2.2uF capacitor to stabilize output voltage. Voltage range is -6.0V~-4.5V
25	VSP	Positive power supply for source driver and step-up circuits. Connect a 2.2uF capacitor to stabilize output voltage. Voltage range is 4.5V~6.0V
26	NC	DUMMY
27	TE	Tearing effect output pin to synchronize MCU to frame writing, activated by S/W command.
28	NC	DUMMY
29	LEDK	Backlight negative
30	LEDA	Backlight positive

3. ABSOLUTE MAXIMUM RATINGS :

Item	Symbol	MIN	TYP	MAX	Unit
Operation Temperature	TOP	-20	-	70	°C
Storage Temperature	TST	-30	-	80	°C

4. ELECTRICAL CHARACTERISTICS

Item	Symbol	MIN	TYP	MAX	Unit
Logic operating voltage	IOVCC	1.65	1.8	3.3	V
Analog operating voltage	VCC	2.4	2.8	3.3	V
Input voltage 'H' level	VIH	0.7IOVCC	-	IOVCC	V
Input voltage 'L' level	VIL	0	-	0.3IOVCC	V
Output voltage 'H' level	VOH	0.8IOVCC	-	-	V
Output voltage 'L' level	VOL	0	-	0.2IOVCC	V

5. BACKLIGHT CHARACTERISTICS

Item	Symbol	Min	Typ	Max	Unit
Working current	I _{BL}	-	40	-	mA
Working voltage	V _{BL}	-	18.5	-	V

6. OPTICAL CHAYACTERISTICS

Item		Symbol	Condition	Min	Typ	Max	Unit	Remark
View Angles		θT	CR≥10	80	85	-	Degree	Note 2
		θB		80	85	-		
		θL		80	85	-		
		θR		80	85	-		
Contrast Ratio		CR	θ=0°	-	1200	-	-	Note 1 Note 3
Response Time		Ton	Viewing normal angle qx=qy=0°	-	25	35	ms	Note 1 Note 4
		Toff						
Chromaticity	White	x	Viewing normal angle qx=qy=0°	-	0.307	-	-	Note 1
		y		-	0.338	-		
	Red	x		-	-	-		
		y		-	-	-		
	Green	x		-	-	-		
		y		-	-	-		
	Blue	x		-	-	-		
		y		-	-	-		
NTSC		-	-	-	55	60	%	
Transmittance		T	-	-	4.3	-	%	Note 1
LCM Brightness		L	Φ=0	-	720	-	cd/m²	Note 5
Brightness Uniformity		BU	Φ=0	80	-	-	%	Note 6

6.1 Measuring Condition:

Measuring surrounding: dark room

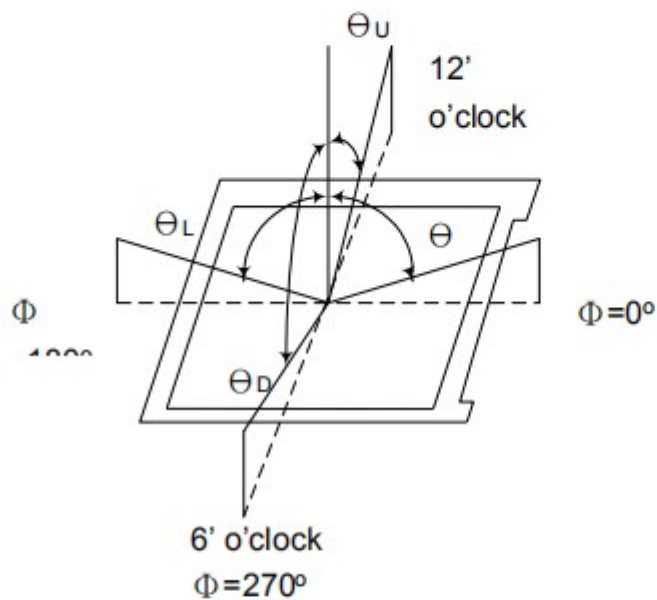
Ambient temperature: $25 \pm 2^\circ\text{C}$

15min. warm-up time.

6.2 Measuring Equipment:

FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics.

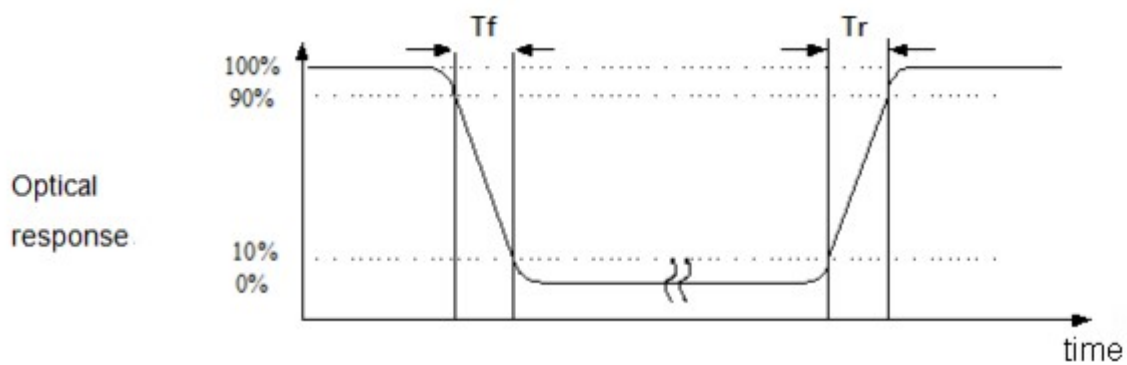
Note (1) Definition of Viewing Angle:



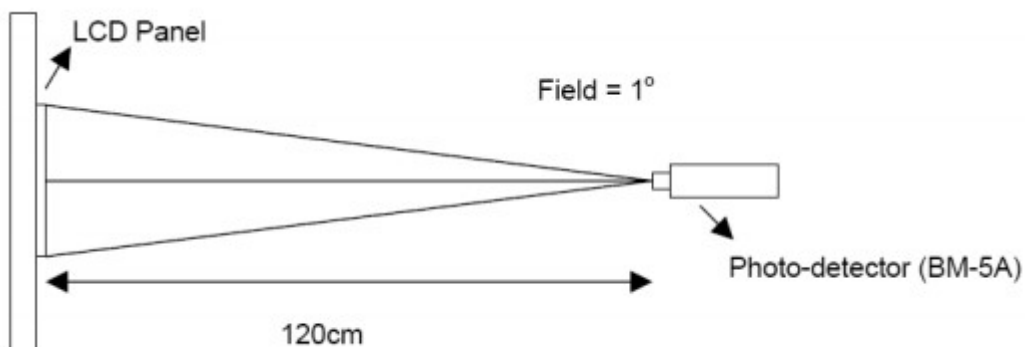
Note (2) Definition of Contrast Ratio (CR) :
measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note (3) Definition of Response Time : Sum of T_R and T_F



Note (4) Definition of optical measurement setup



Note (5)

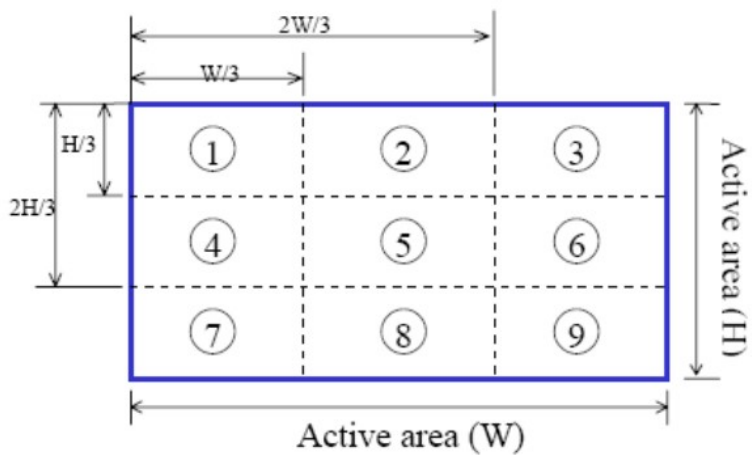
Test condition is:

- (a) Center point on active area
- (b) Best Contrast

Note (6)

Uniformity measure condition:

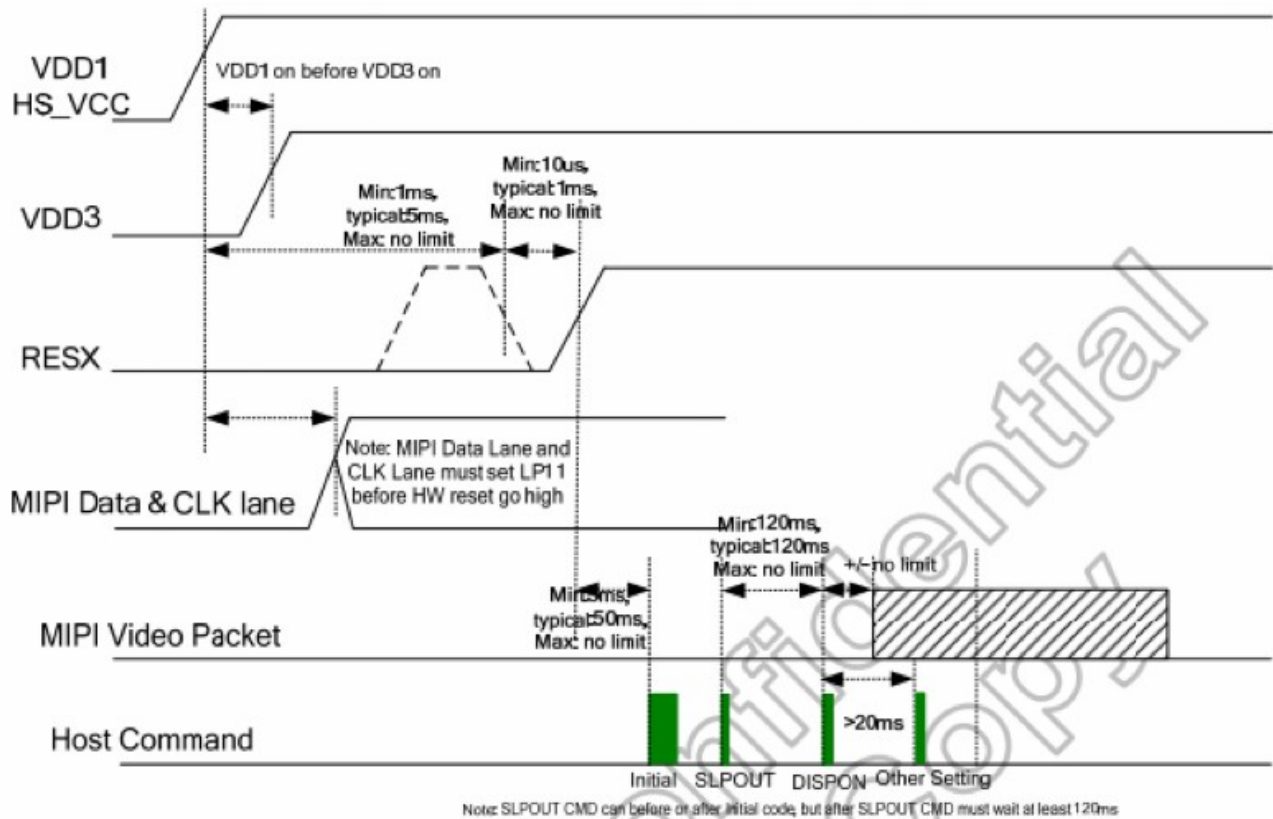
- (1) Measure 9 points. Measure location is shown as below:
- (2) $\text{Uniformity} = (\text{Min. brightness} / \text{Max. brightness}) * 100\%$
- (3) Best Contrast.



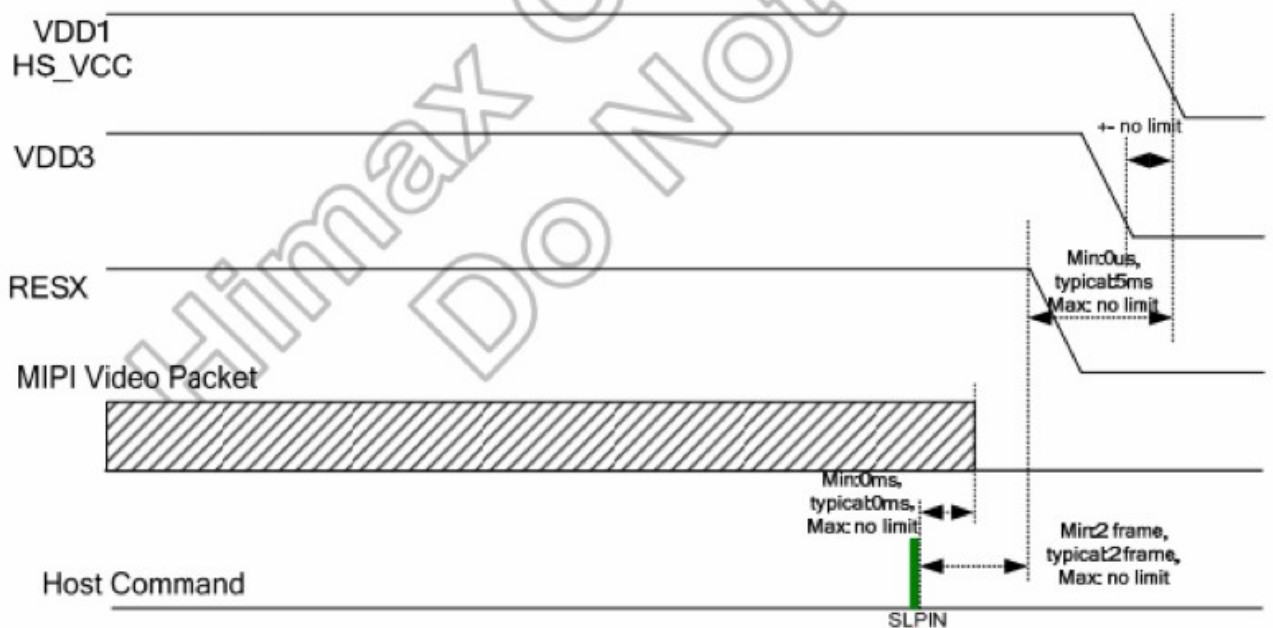
7. TIMING CHARACTERISTIC

Reference to IC datasheet.

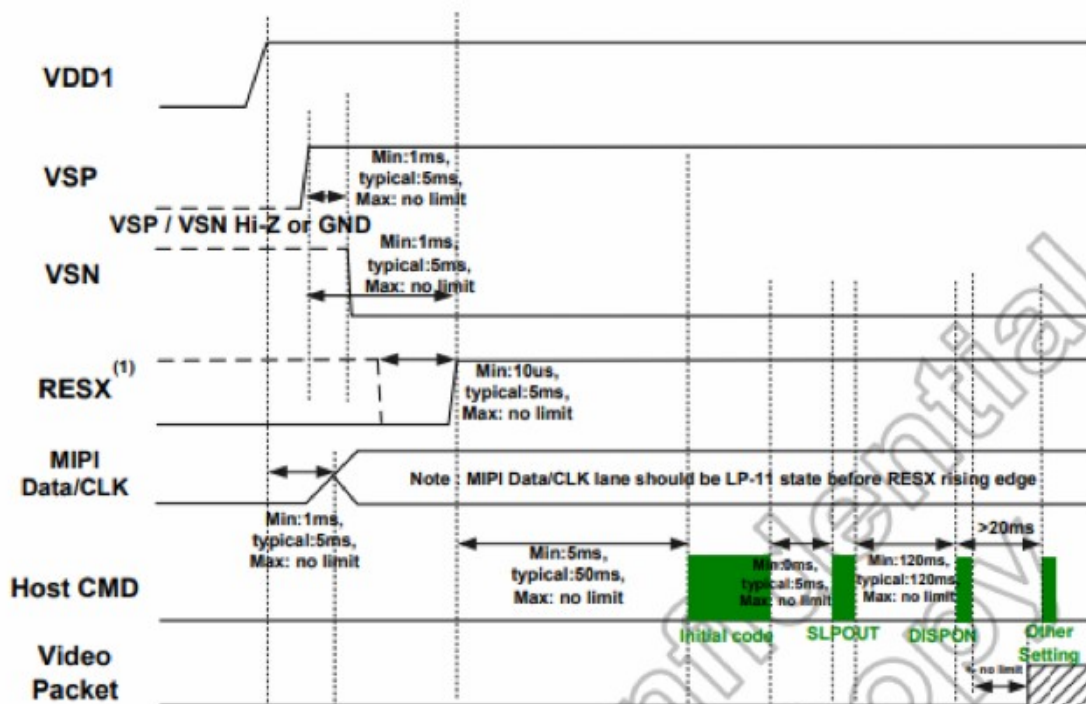
8. POWER ON/OFF SEQUENCE



VDD1/VDD3 input power on sequence

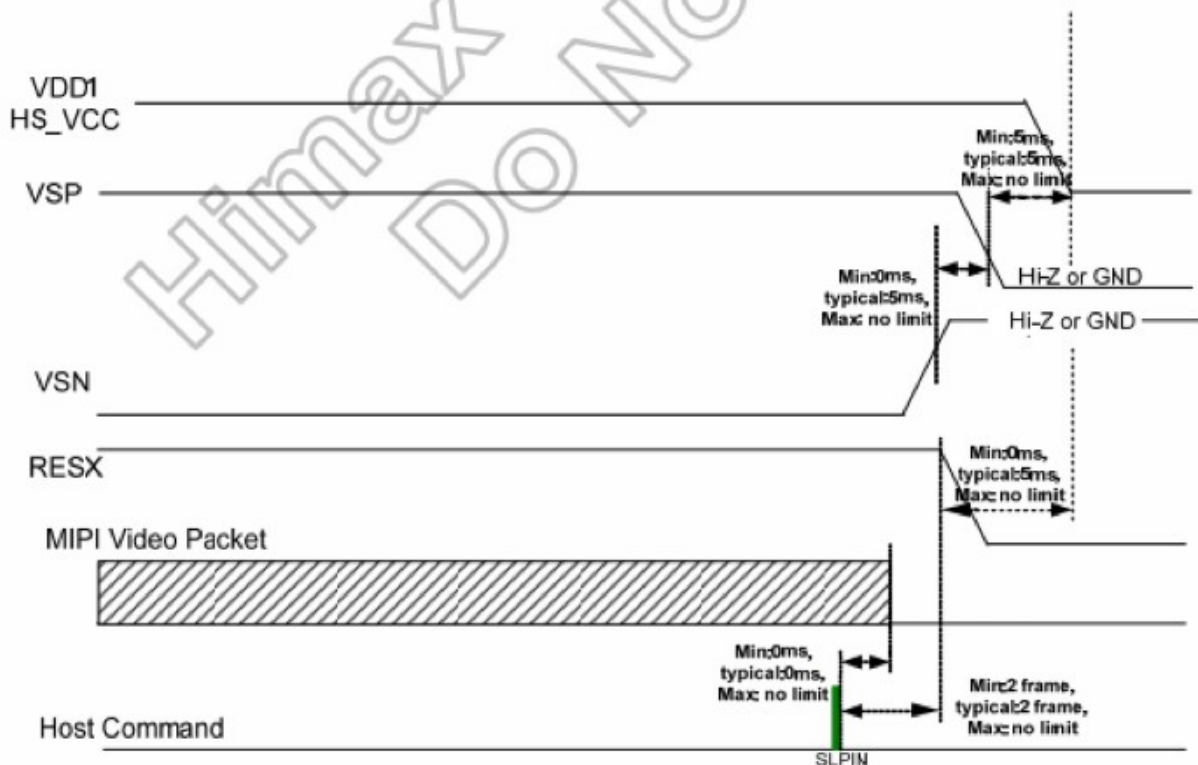


VDD1/VDD3 input power off sequence



VDD1/ VSP/VSN input power on sequence

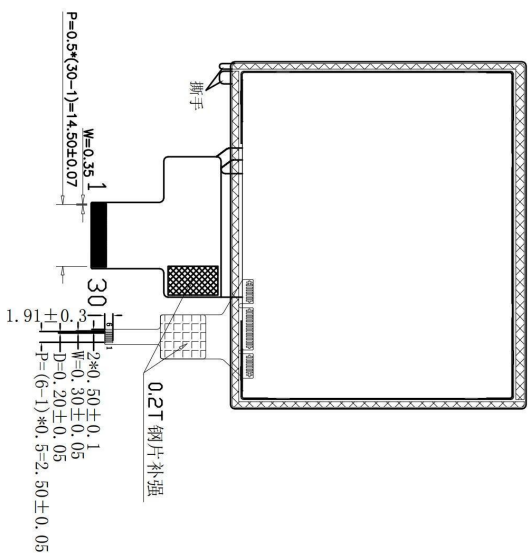
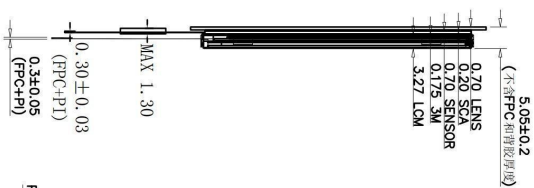
Note1: First RESX rising edge must be after VSP, delay of 1ms or more is recommended. If RESX rising edge comes before VSP, IC might enter un-known state and cause un-expected power on fail due to logic power not ready.



VDD1/VSP/VSN input power off sequence

9. STANDARD SPECIFICATION FOR RELIABILITY

NO.	Test Item	Test Condition	Inspection after test
1	High Temperature Storage	70°C±2°C/240 hours	Inspection after 2~4 hours storage at room temperature, the sample shall be free from below defects: Below defects are not allowed. 1.Air bubble in the LCD 2.Seal leak 3.Non-display 4.Glass crack 5.Current IDD is twice higher than initial value.
2	Low Temperature Storage	-20°C±2°C/120 hours	
3	High Temperature Operating	60°C±2°C/120 hours	
4	Low Temperature Operating	-10°C±2°C/120 hours	
5	High Temperature & High Humidity	50°C, 90% RH/120 hours	
6	Temperature Cycle (Non-operation)	-30°C±2°C(30min.) ~25°C(5min.) ~80°C±2°C(30min.) 20 cycles	
7	ESD Test (Non-operation)	Terminals: ±300 V, Contact Panel: ±6 KV, Air	Sample function is normal.

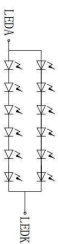


技术参数

1. 结构: 7层板
2. 盖玻: 盖玻片
3. IC及焊盘: IC01510M
4. 工作温度: -20°C ~ +70°C
5. 储存温度: -30°C ~ +80°C
6. 密封腔测试: 40kPa球, 40cm高度中心点, 无泄漏情况
7. 硬度要求: CS: ≤ 150 nm, DOL: ≤ 35 nm, 40P ≤ 550 nm, 翘曲度: ≤ 25 mm
8. 表面处理: $\geq 6H$ Glass
9. 环保: RoHS
10. 透光率: 85%
11. 厚度: 3mm
12. 未标尺寸按±0.2mm, 开孔尺寸公差±0.05mm(置公差: 0.15)

- 1、Display type: 3.5"-inch
- 2、Viewing direction: ALL
- 3、Drive IC: HX8394F
- 4、Backlight color: 12 LED WHITE
- 5、Operation temperature: -20~70° C
- 6、Storage temperature: -30~80° C
- 7、LCM Operating Voltage: Vdd=2.8V
- 8、Vands: 模块可视区域, 建议外壳可视

VF=18.5V IF=40MA



展开出货

			DATE	2025. 05. 05
1 : 1	UNIT	mm	APPROVED	
			CHECKED	
			DESIGNED	

11. INCOMING INSPECTION SPECIFICATION

11.1 Inspection conditions:

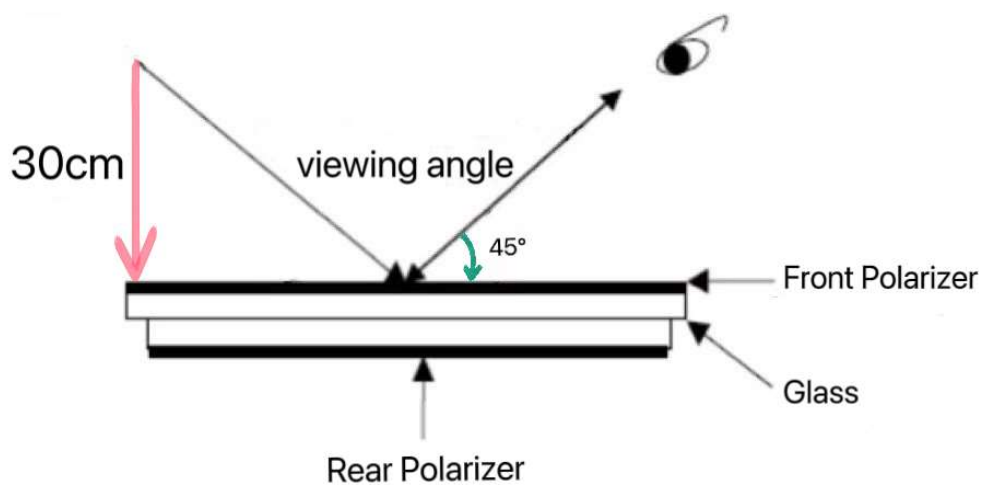
Temperature: 25+/- 5°C

Humidity: 65%+/-10%RH

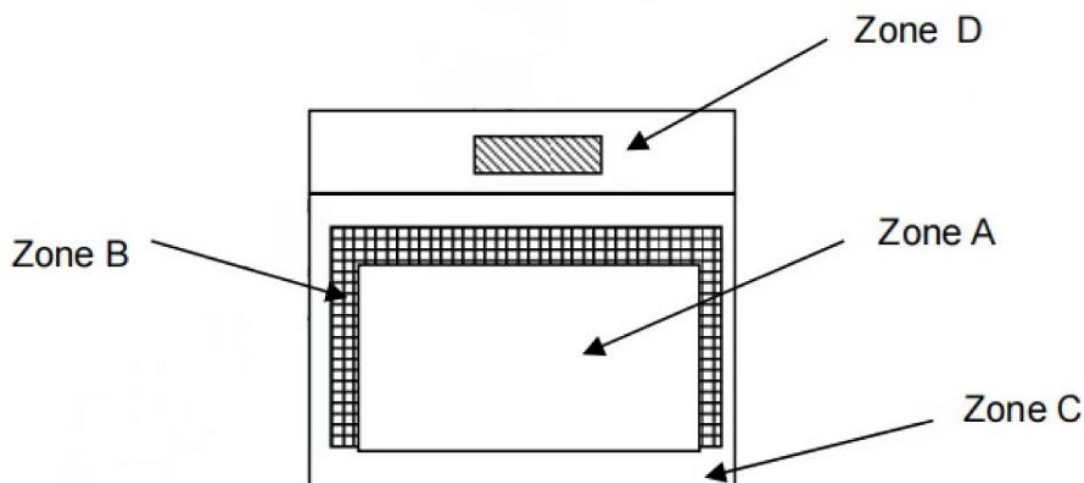
Ambient illumination: 300+/-50 Lux

Viewing distance: 30cm

Viewing angle: within 45 ° to the perpendicular line.



11.2 Definition of Zone



Zone A : Effective Viewing Area(Character or Digit can be seen)

Zone B : Viewing Area except Zone A

Zone C : Outside (Zone A+Zone B) which can not be seen after assembly by customer .

Zone D : IC Bonding Area

Note:As a general rule ,visual defects in Zone C can be ignored when it doesn't effect product function or appearance after assembly by customer

11.3 Definition of dot defect

- (a) The definition of dot: The size of a defective dot over 1/2 of whole dot is regarded as one defective dot, and all brightness or dark dot defect must be visible through ND 5% filter.
- (b) Bright dot: Dots appear bright and unchanged in size in which LCD Cell is displaying under black pattern.
- (c) Dark dot: Dots appear dark and unchanged in size in which LCD Cell is displaying under pure red, green, blue picture.
- (d) 2 dot adjacent=1 pair=2 dots



2 dot adjacent



2 dot adjacent



2 dot adjacent(vertical)



2 dot adjacent(slant)

11.4 Sampling Plan

According to GB/T 2828-2003 ; , normal inspection, Class II

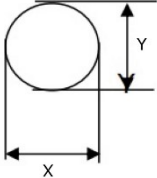
AQL:

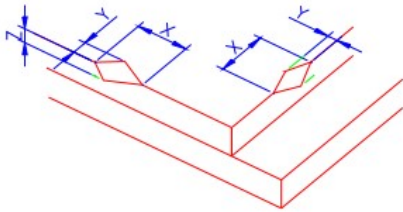
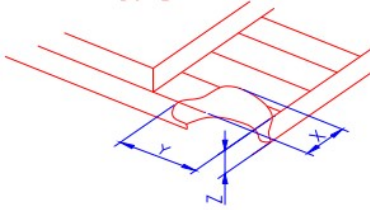
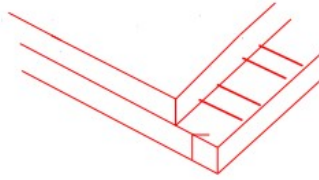
Major defect	Minor defect
0.4	0.65

LCD: Liquid Crystal Display , TP: Touch Panel , LCM: Liquid Crystal Module

No	Items to be inspected	Criteria	Classification of defects
1	Functional defects	1) No display, Open or miss line 2) Display abnormally, Short 3) Backlight no lighting, abnormal lighting. 4) TP no function	Major
2	Missing	Missing component	
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed	
4	Color tone	Color unevenness, refer to limited sample	Minor
5	Spot/Linear defect	Light dot, Dim spot, Polarizer Bubble ; Polarizer accidented spot.	
6	Soldering appearance	Good soldering , Peeling off is not allowed.	
7	LCD/Polarizer/TP	Black/White spot/linear, scratch, crack, etc.	

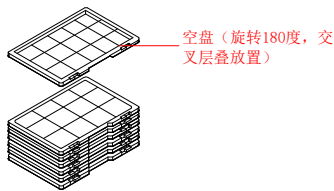
11.5 Criteria(visual)

Number	Items	Criteria(mm)																																																																																																										
1.0	Spot defect  $\Phi=(X+Y)/2$	① Light spot（LCD/TP/Polarizer/BL white spot , pinhole,dent, stain） <table><tr><th>Zone</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>Size(mm)</th><th>A</th><th>B</th><th>C</th></tr><tr><td>$\Phi\leq0.10$</td><td colspan="3">Ignore</td></tr><tr><td>$0.10<\Phi\leq0.25$</td><td colspan="3">3(distance $\geq 10\text{mm}$)</td></tr><tr><td>$0.25<\Phi\leq0.30$</td><td colspan="3">1</td></tr><tr><td>$\Phi>0.30$</td><td colspan="3">0</td></tr></table> ②Dim spot（LCD/TP/Polarizer /BL/dim dot, light leakage、 dark spot） <table><tr><th>Zone</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>Size(mm)</th><th>A</th><th>B</th><th>C</th></tr><tr><td>$\Phi\leq0.10$</td><td colspan="3">Ignore</td></tr><tr><td>$0.10<\Phi\leq0.25$</td><td colspan="3">3(distance $\geq 10\text{mm}$)</td></tr><tr><td>$0.25<\Phi\leq0.30$</td><td colspan="3">1</td></tr><tr><td>$\Phi>0.30$</td><td colspan="3">0</td></tr></table> ③ Polarizer accidented spot <table><tr><th>Zone</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>Size(mm)</th><th>A</th><th>B</th><th>C</th></tr><tr><td>$\Phi\leq0.20$</td><td colspan="3">Ignore</td></tr><tr><td>$0.20<\Phi\leq0.30$</td><td colspan="3">2(distance $\geq 10\text{mm}$)</td></tr><tr><td>$\Phi>0.30$</td><td colspan="3">0</td></tr></table> ④Pixel defective dots（Bright dots, Dark dots） <table><tr><th>Zone</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>Size(mm)</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Bright dots</td><td>1</td><td colspan="2" rowspan="3">Ignore</td></tr><tr><td>Dark dots</td><td>2(distance $\geq 10\text{mm}$)</td></tr><tr><td>Total</td><td>3</td></tr></table> ⑤ Polarizer Bubble <table><tr><th>Zone</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>Size(mm)</th><th>A</th><th>B</th><th>C</th></tr><tr><td>$\Phi\leq0.20$</td><td colspan="3">Ignore</td></tr><tr><td>$0.20<\Phi\leq0.30$</td><td>1</td><td>3</td><td rowspan="3">Ignore</td></tr><tr><td>$0.30<\Phi\leq0.40$</td><td>1</td><td>2(distance $\geq 10\text{mm}$)</td></tr><tr><td>$\Phi>0.40$</td><td colspan="2">0</td></tr></table>	Zone	Acceptable Q'ty			Size(mm)	A	B	C	$\Phi\leq0.10$	Ignore			$0.10<\Phi\leq0.25$	3(distance $\geq 10\text{mm}$)			$0.25<\Phi\leq0.30$	1			$\Phi>0.30$	0			Zone	Acceptable Q'ty			Size(mm)	A	B	C	$\Phi\leq0.10$	Ignore			$0.10<\Phi\leq0.25$	3(distance $\geq 10\text{mm}$)			$0.25<\Phi\leq0.30$	1			$\Phi>0.30$	0			Zone	Acceptable Q'ty			Size(mm)	A	B	C	$\Phi\leq0.20$	Ignore			$0.20<\Phi\leq0.30$	2(distance $\geq 10\text{mm}$)			$\Phi>0.30$	0			Zone	Acceptable Q'ty			Size(mm)	A	B	C	Bright dots	1	Ignore		Dark dots	2(distance $\geq 10\text{mm}$)	Total	3	Zone	Acceptable Q'ty			Size(mm)	A	B	C	$\Phi\leq0.20$	Ignore			$0.20<\Phi\leq0.30$	1	3	Ignore	$0.30<\Phi\leq0.40$	1	2(distance $\geq 10\text{mm}$)	$\Phi>0.40$	0	
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$\Phi>0.40$	0																																																																																																											

2.0	Linear defect (LCD/TP /Polarizer /backlight black/white line, scratch, stain)	<table><tr><th rowspan="2">Width(mm)</th><th rowspan="2">Length(mm)</th><th colspan="3">Acceptable Q'ty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>$\Phi\leq0.03$</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr><tr><td>$0.03<W\leq0.04$</td><td>$L\leq3.0$</td><td colspan="2">$N\leq2$</td></tr><tr><td>$0.04<W\leq0.06$</td><td>$L\leq2.0$</td><td colspan="2">$N\leq2$</td></tr><tr><td>$0.06<W$</td><td colspan="4">Define as spot defect</td></tr></table>	Width(mm)	Length(mm)	Acceptable Q'ty			A	B	C	$\Phi\leq0.03$	Ignore	Ignore		Ignore	$0.03<W\leq0.04$	$L\leq3.0$	$N\leq2$		$0.04<W\leq0.06$	$L\leq2.0$	$N\leq2$		$0.06<W$	Define as spot defect			
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$0.04<W\leq0.06$	$L\leq2.0$	$N\leq2$																										
$0.06<W$	Define as spot defect																											
3.0	Electronic Components SMT	Not allow missing parts, solderless connection, cold solder joint, mis match, The positive and negative polarity opposite																										
4.0	Display color& Brightness	1. Color : Measuring the color coordinates, The measurement standard according to the datasheet or samples. 2. Brightness : Measuring the brightness of White screen, The measurement standard according to the datasheet or Samples.																										
5.0	LCD Mura	By 5% ND filter invisible.																										
6.0 LCD Crack/Broken NOTE: X: Length Y: Width Z: Height L:Length of ITO T:Height of LCD	(1) The edge of LCD broken	 <table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>$\leq3.0\text{mm}$</td><td><Inner border line of the seal</td><td>$\leq T$</td></tr></table>	X	Y	Z	$\leq3.0\text{mm}$	<Inner border line of the seal	$\leq T$																				
	X	Y	Z																									
	$\leq3.0\text{mm}$	<Inner border line of the seal	$\leq T$																									
(2)LCD corner broken	 <table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>$\leq3.0\text{mm}$</td><td>$\leq L$</td><td>$\leq T$</td></tr></table>	X	Y	Z	$\leq3.0\text{mm}$	$\leq L$	$\leq T$																					
X	Y	Z																										
$\leq3.0\text{mm}$	$\leq L$	$\leq T$																										
(3) LCD crack	 Crack Not allowed																											

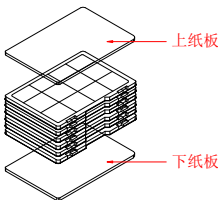
12. PACKING SPEC

12.1 Packing drawing

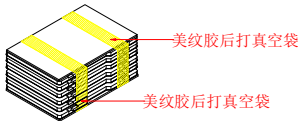


空盘（旋转180度，交叉层叠放置）

步骤一



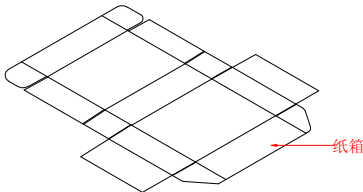
步骤二



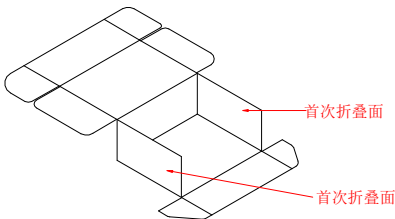
美纹胶后打真空袋

美纹胶后打真空袋

步骤三



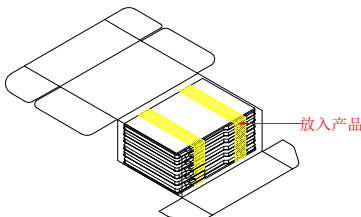
步骤四



首次折叠面

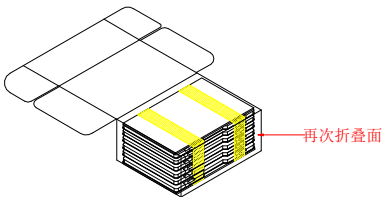
首次折叠面

步骤五



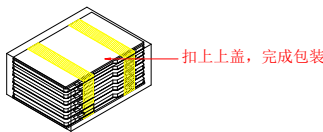
放入产品

步骤六



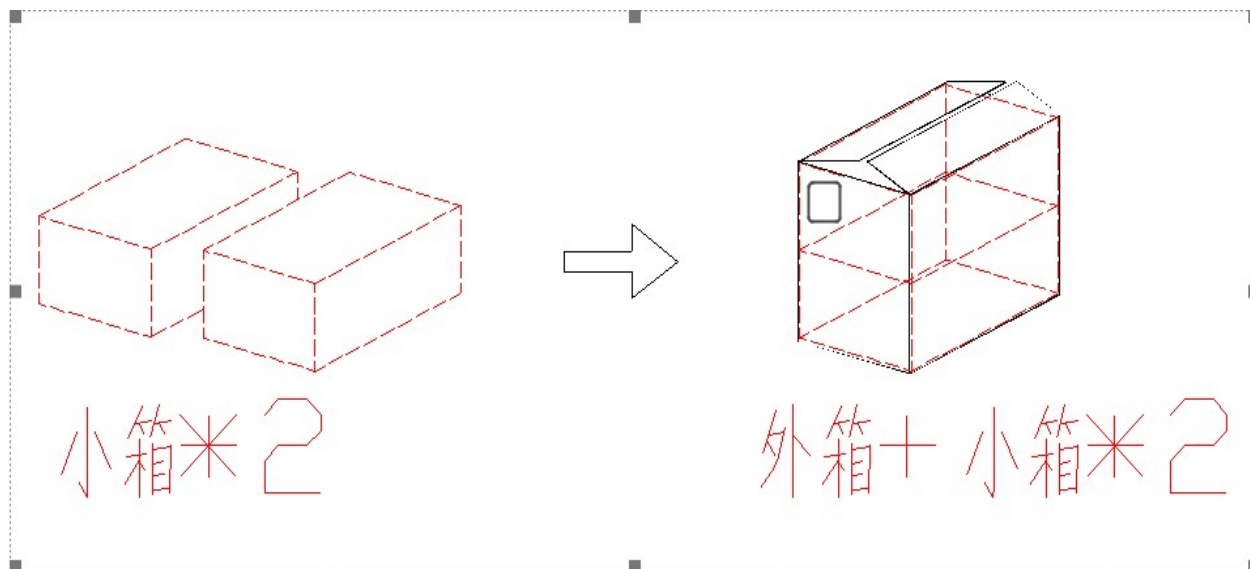
再次折叠面

步骤七



扣上上盖，完成包装

步骤八



NO.	Item	Dimensions	Quantity	Remark
1	TRAY	Tray		
2	Inner Carton	One carton(Including trays)		
3	Outer Carton	One carton(Including 2 Inner cartons)	1	

12.2 LCD Module Label(printing) information

ET35176BVIC
Date
Made in China

And the Part Number definition is as below:

ET Company Name
35 Panel Size
176 LCM model code(Internal control)
B Panel maker
V Resolution
I Connector type
C Version

12.3 Carton Label information:

Customer:
Name of material: LCD Module
Part Number: ET35176BVIC
Quantity:
Made in China
Date:

13.CAUTIONS AND HANDLING PRECAUTIONS

13.1 Handling and Operating the Module

- (1) When the module is assembled, it should be attached to the system firmly. Do not warp or twist the module during assembly work.
- (2) Protect the module from physical shock or any force. In addition to damage, this may cause improper operation or damage to the module and back-light unit.
- (3) Note that polarizer is very fragile and could be easily damaged. Do not press or scratch the surface.
- (4) Do not allow drops of water or chemicals to remain on the display surface. If you have the droplets for a long time, staining and discoloration may occur.
- (5) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.
- (6) The desirable cleaners are water, IPA (Isopropyl Alcohol) or Hexane. Do not use ketene type materials (ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanent damage to the polarizer due to chemical reaction.
- (7) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth. In case of contact with hands, legs, or clothes, it must be washed away thoroughly with soap.
- (8) Protect the module from static; it may cause damage to the CMOS ICs.
- (9) Use finger-stalls with soft gloves in order to keep display clean during the incoming inspection and assembly process.
- (10) Do not disassemble the module.
- (11) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.
- (12) Pins of I/F connector shall not be touched directly with bare hands.
- (13) Do not connect, disconnect the module in the “Power ON” condition.
- (14) Power supply should always be turned on/off by the item Power On Sequence & Power Off Sequence.

13.2 Storage and Transportation.

- (1) Do not leave the panel in high temperature, and high humidity for a long time.
It is highly recommended to store the module with temperature from 0 to 35 °C and relative humidity of less than 70%
- (2) Do not store the TFT-LCD module in direct sunlight.
- (3) The module shall be stored in a dark place. When storing the modules for a long time, be sure to adopt effective measures for protecting the modules from strong ultraviolet radiation, sunlight, or fluorescent light.
- (4) It is recommended that the modules should be stored under a condition where no condensation is allowed. Formation of dewdrops may cause an abnormal operation or a failure of the module. In particular, the greatest possible care should be taken to prevent any module from being operated where condensation has occurred inside.
- (5) This panel has its circuitry FPC on the bottom side and should be handled carefully in order not to be stressed.